Design and Implementation of an SDR-Based Wireless Channel Sounder for Indoor Multipath Characterization

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UCSD WES 207

Motivation / Problem

Indoor wireless channels exhibit:

Multipath reflections

Delay spread

Frequency-selective fading

Rapid spatial variation

Applications:

Wi-Fi

5G

radar

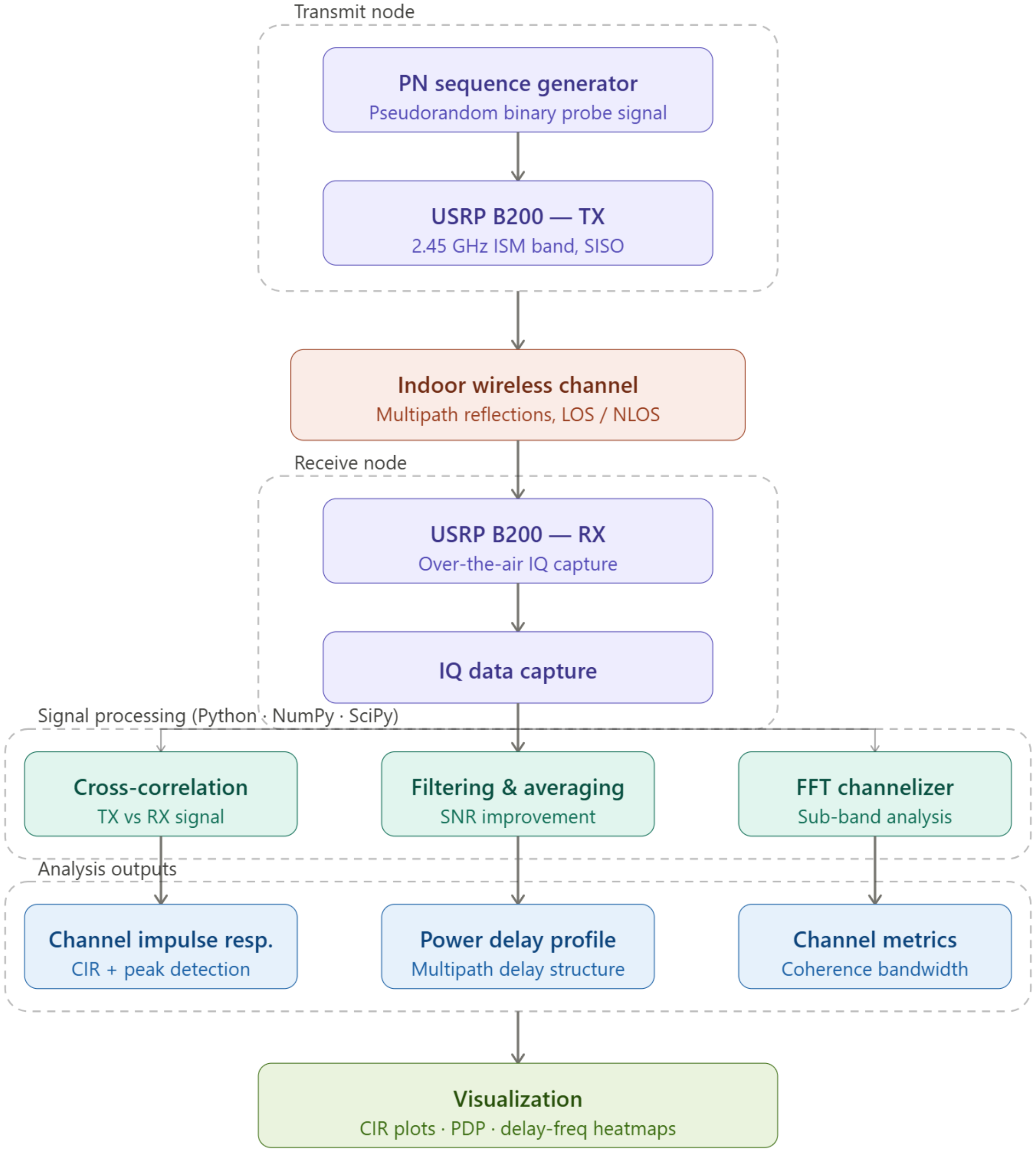
tactical communications

adaptive RF systems

Commercial channel sounders are expensive and difficult to modify.

Goal: build a flexible SDR-based propagation analysis platform.

System Architecture



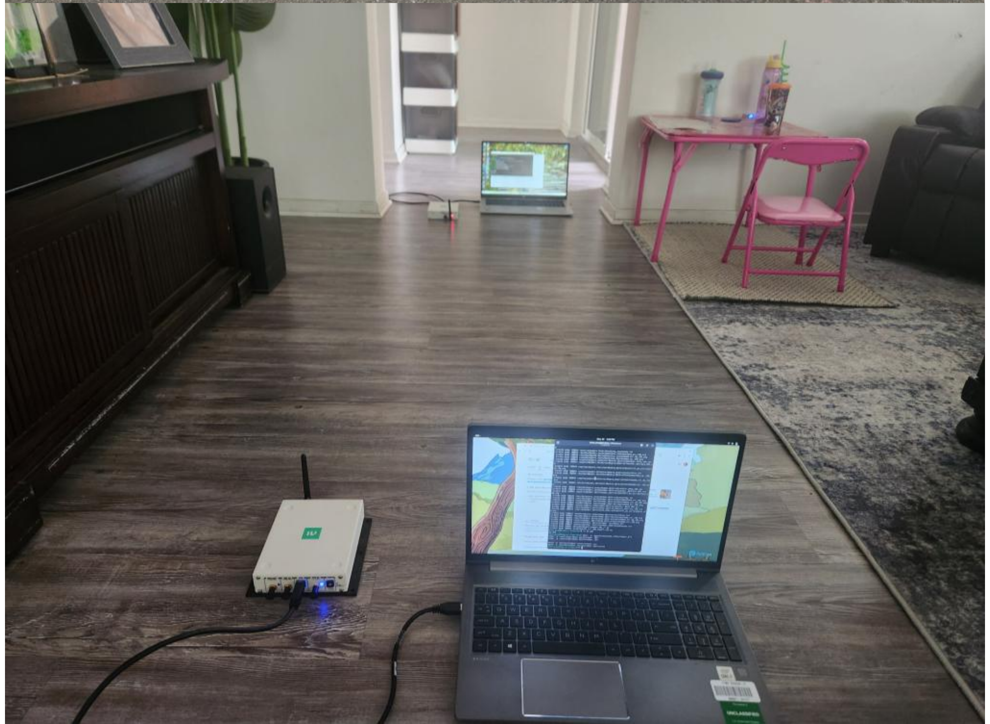
Hardware + Experimental Setup

2× Ettus USRP B200 SDRs

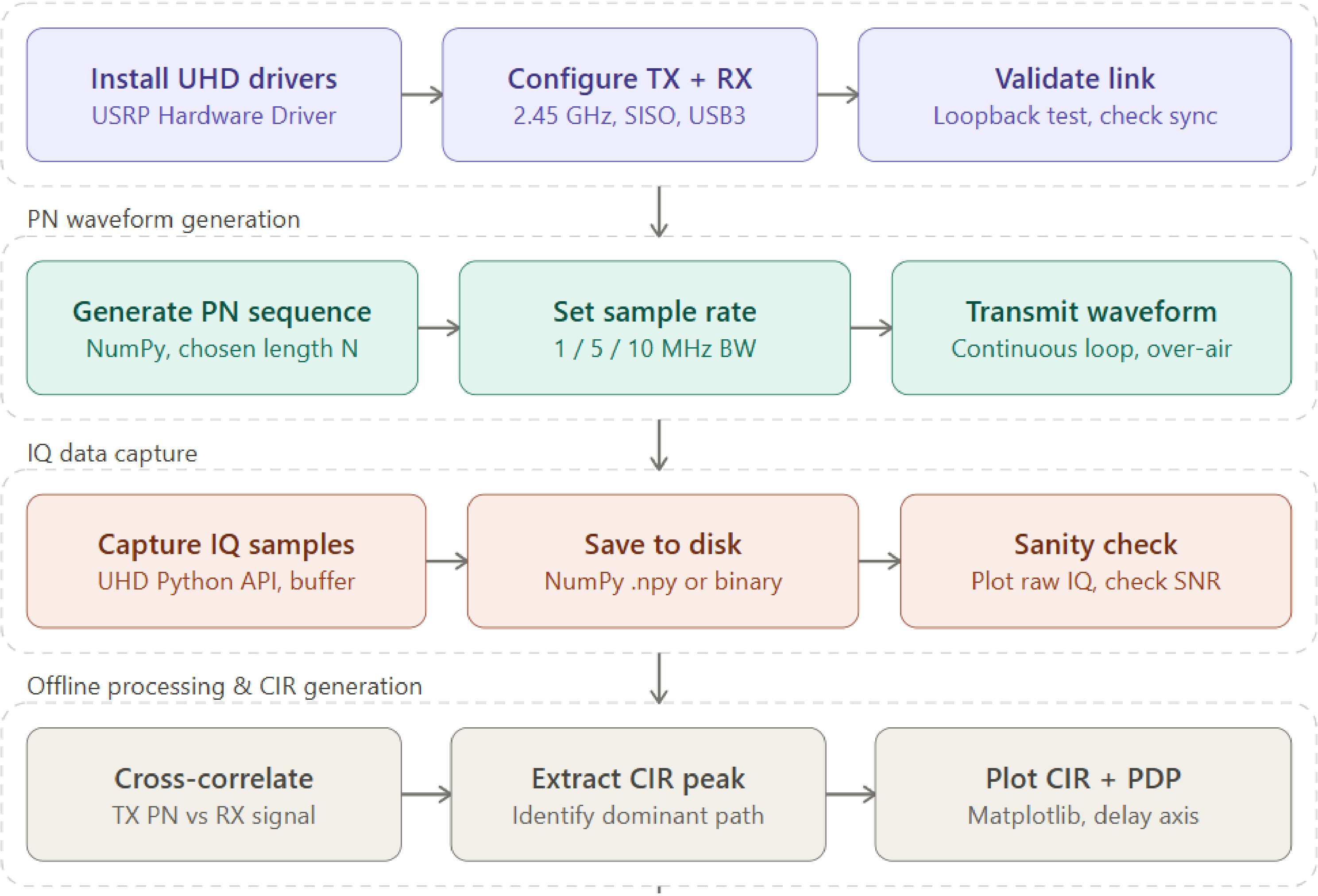
Python + UHD API

2.45 GHz ISM band

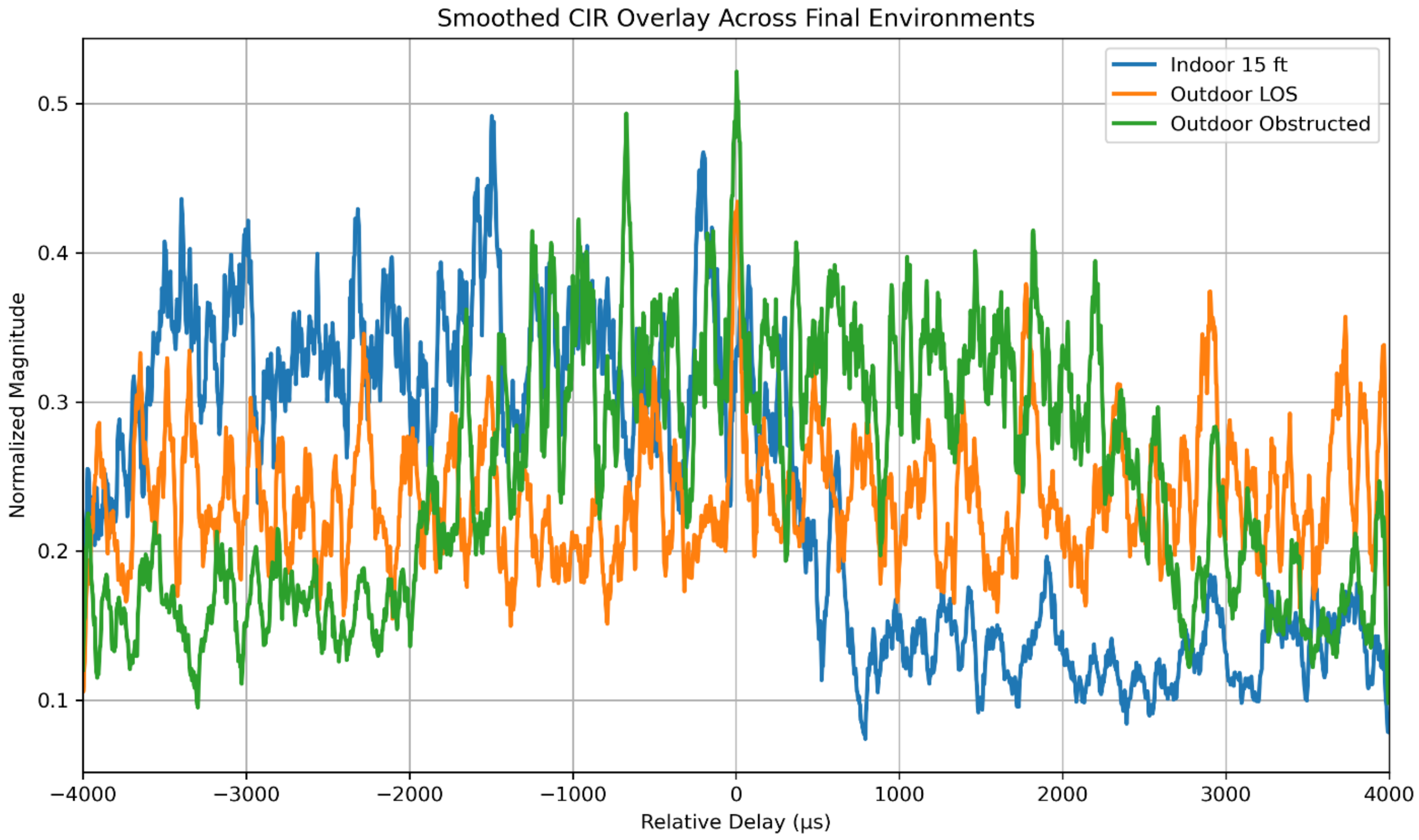
Offline IQ capture and processing



Signal Processing Workflow

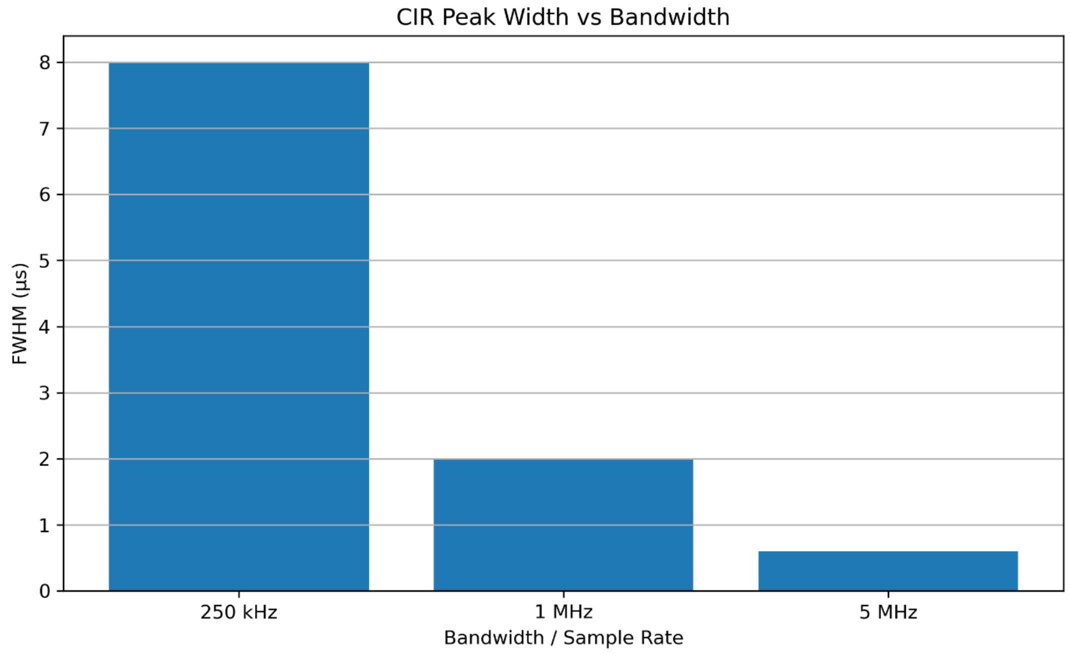


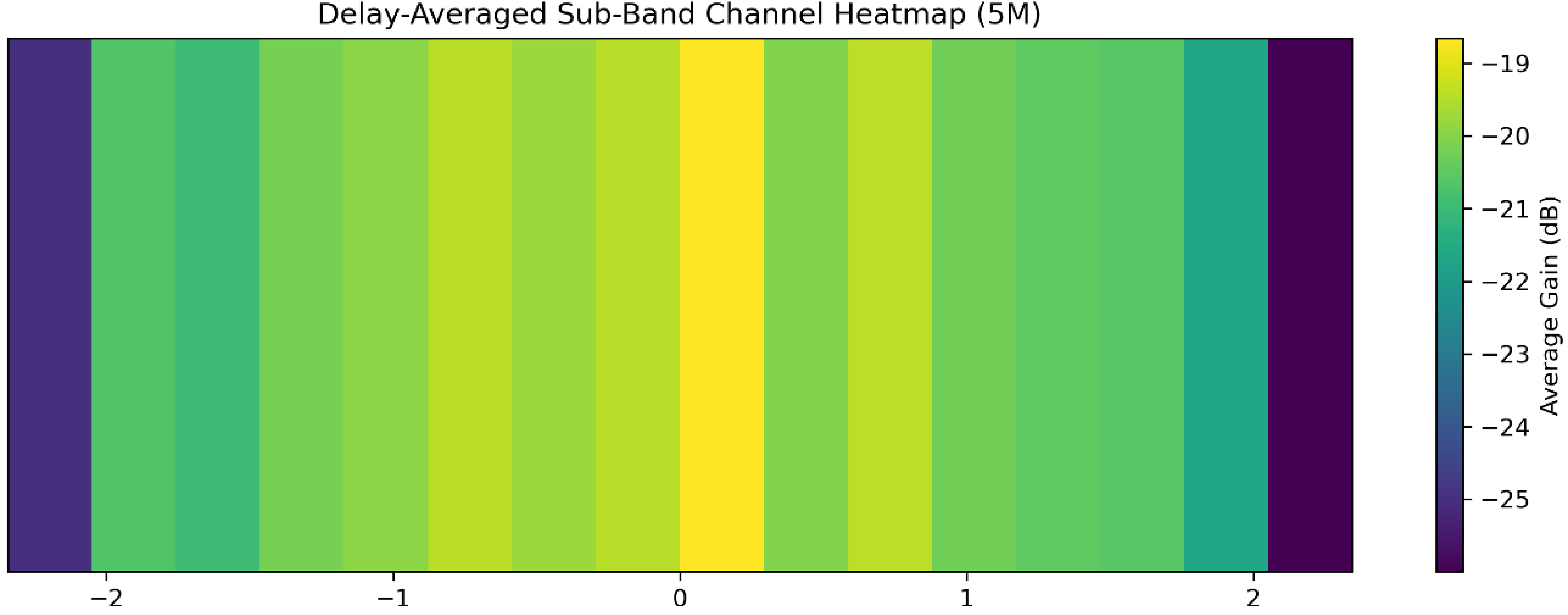
Core Results



Bandwidth Results

- Higher bandwidth improved temporal resolution.
- 250 kHz → ~8 µs
- 1 MHz → ~2 µs
- 5 MHz → ~0.6 µs





Frequency Selective Analysis

- FFT-based channelizer enabled sub-band propagation analysis
- Visualized frequency-selective fading across frequency

Key Lessons Learned

SDR SYNCHRONIZATION IS DIFFICULT

INDOOR MULTIPATH IS HIGHLY SPATIALLY SENSITIVE

OFFLINE PROCESSING GREATLY SIMPLIFIED DEBUGGING

VISUALIZATION IS CRITICAL FOR INTERPRETING RF BEHAVIOR

Conclusion / Future Work

Achieved:

end-to-end SDR channel sounding

CIR/PDP extraction

environmental comparison

FFT-based analysis

Future:

MIMO

Doppler analysis

Real-time processing

Phase-coherent synchronization

Thank You

GitHub

Scan QR Code for Full Project Repository